# Byungjun Kim (김병준)

Assistant Professor of Cultural Informatics, Graduate School of Korean Studies,
The Academy of Korean Studies

bjkim@byungjunkim.com · bjkim@aks.ac.kr · +82-31-730-8161

byungjunkim.com · Lab: digitalhumanities.kr · Google Scholar · ORCID 0000-0001-9925-9343

## RESEARCH PROFILE

My research lies in digital humanities and computational social science: I analyze large-scale bibliographic and textual data to study how scholarly and cultural knowledge is produced and communicated.

**Keywords:** digital humanities · computational social science · cultural analytics · bibliometrics and science of science · natural language processing and text mining · scholarly communication and open access · Korean datasets and language models

## EDUCATION

| | |
|---|---|
| Mar. 2018 – Aug. 2022 | **Ph.D. in Data Science**, Sungkyunkwan University (Dept. of Interaction Science)<br>Dissertation: *Mapping the Knowledge Structure of Korean Humanities: Bibliographic Data Analysis of Humanities Journal Articles in the Korea Citation Index, 2004–2019*. Advisor: Jang Hyun Kim. |
| Mar. 2015 – Feb. 2017 | **M.S. in Culture Technology**, KAIST (Graduate School of Culture Technology)<br>Thesis: *The Shift of Critical Language: The Shifting Aspects of Concept Words in Literature-Magazine Criticism, 1995–2015*. Advisors: Wonjae Lee and Bong Gwan Jun. |
| Mar. 2008 – Feb. 2015 | **B.A. in Korean Language and Literature**, Sungkyunkwan University (dual major: Film, Television and Multimedia) |

## ACADEMIC & RESEARCH APPOINTMENTS

| | |
|---|---|
| Sep. 2024 – Present | **Assistant Professor**, Cultural Informatics, Graduate School of Korean Studies, The Academy of Korean Studies |
| Sep. 2024 – Present | **Co-Director**, Center for Digital Humanities, The Academy of Korean Studies |
| Jan. 2023 – Aug. 2024 | **Research Assistant Professor**, Center for Digital Humanities & Computational Social Sciences, KAIST |
| Mar. 2024 – May 2024 | **Research Fellow**, Trier Center for Digital Humanities, Universität Trier, Germany (CLS INFRA) |
| Nov. 2021 – Dec. 2022 | **Research Associate**, Center for Digital Humanities & Computational Social Sciences, KAIST |
| Jan. 2018 – Nov. 2021 | **Graduate Research Assistant**, Data Science & Social Analytics Lab (DSSAL), Sungkyunkwan University |
| Jun. 2017 – Dec. 2018 | **Manager**, Data Research Team, Bigvalue (real estate data analysis) |

## PEER-REVIEWED INTERNATIONAL JOURNAL ARTICLES (SSCI · A&HCI · SCIE · SCOPUS)

*Underline indicates corresponding authorship throughout this CV; * indicates co-first authors (equal contribution); † indicates co-corresponding authors.*

1. Kim, G., Park, S., Ji, H., Lee, H., Lee, B., Jeong, C., & **Kim, B.** (2026). KNoTE: A TEI encoded XML dataset of modern Korean novels. *Journal of Open Humanities Data*, 12, 102. doi.org/10.5334/johd.590
2. Lim, I., Ji, H., & **Kim, B.** (2026). KPoEM: A human-annotated dataset for emotion classification and RAG-based poetry generation in Korean modern poetry. *The Review of Korean Studies*, 29(1), 161–206. 10.25024/review.2026.29.1.006
3. **Kim, B.**, Jee, J., & Woo, D. (2026). Mapping the semantic transformations of major powers in Cold War East Asia: A word embedding approach using Chinese and Korean newspapers (1946–95). *Digital Scholarship in the Humanities*, *41*(2), 778-793. doi.org/10.1093/llc/fqag003
4. Hong, Y., **Kim, B.**, Jeon, J., & Kim, L. (2025). Has higher education become more interdisciplinary? A longitudinal analysis of syllabi using natural language processing. *Humanities and Social Sciences Communications*, 12(1), 1841. doi.org/10.1057/s41599-025-06126-7
5. Lee, S., **Kim, B.**, & Jun, B. G. (2024). Automatic detection of four-panel cartoon in large-scale Korean digitized newspapers using deep learning. *Journal of Open Humanities Data*, 10, 36. doi.org/10.5334/johd.205
6. Jee, J., **Kim, B.**, & Jun, B. G. (2024). The role of English Wikipedia in mediating East Asian historical disputes: The case of Balhae. *Asian Journal of Communication*, 34(4), 399-418. doi.org/10.1080/01292986.2024.2342822
7. **Kim, B.***, Yang, S.*, & Kim, H. (2024). Voices of transitions: Korea's online news media and user comments on the energy transition. *Energy Policy*, 187, 114020. doi.org/10.1016/j.enpol.2024.114020
8. Song, J., **Kim, B.**, Kim, M., & Iverson, P. (2023). The Korean Speech Recognition Sentences: A large corpus for evaluating semantic context and language experience in speech perception. *Journal of Speech, Language, and Hearing Research*, 66(9), 3399-3412. doi.org/10.1044/2023_JSLHR-23-00137
9. Kim, Y., **Kim, B.**, Park, M., Nam, W., & Kim, J. H. (2023). A soft power challenge, or an opportunity? A big data analysis on Chinese soft power during the COVID-19 pandemic. *Foreign Policy Analysis*, 19(3), orad011. doi.org/10.1093/fpa/orad011
10. Lee, K. R., Kim, J. H., Jang, J., Yoon, J., Nan, D., Kim, Y., & **Kim, B.** (2023). News big data analysis of international start-up innovation discourses through topic modelling and network analysis: Comparing East Asia and North America. *Asian Journal of Technology Innovation*, 31(3), 581–603. doi.org/10.1080/19761597.2022.2134154
11. Lee, K. R., **Kim, B.**, Nan, D., & Kim, J. H. (2021). Structural topic model analysis of mask-wearing issue using international news big data. *International Journal of Environmental Research and Public Health*, 18(12), 6432. doi.org/10.3390/ijerph18126432
12. **Kim, B.**, Yoo, M., Park, K. C., Lee, K. R., & Kim, J. H. (2021). A value of civic voices for smart city: A big data analysis of civic queries posed by Seoul citizens. *Cities*, 108, 102941. 10.1016/j.cities.2020.102941

## REFEREED INTERNATIONAL CONFERENCE PROCEEDINGS (AI/NLP)

1. van Cranenburgh, A., Yang, X., Alvanita, Di Domenico, C. N., Ferragud, M., Graciotti, A., Ion, A. G., **Kim, B.**, Park, S., Visser Solissa, N., Zhou, X., & Pianzola, F. (2026). GOLEMcoref: A multilingual coreference

dataset of fiction. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 472–480. https://aclanthology.org/2026.acl-short.39/

## PEER-REVIEWED DOMESTIC JOURNAL ARTICLES (KCI)

1. 구슬, 김가연, **김병준**. 「LLM 상용화 이후 한국 인문사회과학 학술 초록의 어휘 변화: 2004~2024년 KCI 초록 데이터 기반 분석」. *언어와 정보*, 30(1): 1–24, 2026. doi.org/10.29403/LI.30.1.1
2. 최주찬, **김병준**, 김경민, 허예슬. 「소설의 시대 31년, 그 백 번째 물음 — 데이터로 읽는 『현대소설연구』 1~99호」. *현대소설연구*, 100, 2025. doi.org/10.20483/JKFR.2025.12.100.9
3. **김병준**, 노대원. 「생성형 AI는 인문학 연구를 어떻게 바꿀까?」. *영주어문*, 59, 2025. doi.org/10.30774/yjll.2025.2.59.203
4. 최진석, 최새흰, **김병준**, 허예슬, 최주찬, 황호덕. 「두 박완서, 제도와 젠더 혹은 디지털 작가 연구의 시좌 — 2010년대의 『여성문학연구』를 중심으로」. *개념과 소통*, 34, 2024. doi.org/10.15797/concom.2024..34.004
5. 서재현, 하나, 김민우, 한지민, **김병준**, 박소정. 「멀리서 읽는 "저희" — ELECTRA 기반 의미 분류 모델과 N-gram을 활용한 두 가지 '저희'의 통시적 변화 양상 연구」. *대동문화연구*, 128, 2024. doi.org/10.18219/ddmh..128.202412.513
6. (KCI 미등재) **김병준**. 「근대 국한문혼용체 자료 서브워드 기반 형태소 분석기의 설계와 적용」. *디지털인문학*, 1(2), 2024. doi.org/10.23287/KJDH.2024.1.2.5
7. (우수등재) 장우리, **김병준**, 전봉관. 「한국 최초 신문연재만화 「멍텅구리」의 서사 구조와 문화사적 의미 연구」. *인문콘텐츠*, 74, 2024. doi.org/10.18658/humancon.2024.09.231
8. 이윤호, **김병준**. 「한국 학술장 내 웹툰 관련 연구의 확장과 분화: KCI 논문 서지 데이터 기반 구조적 토픽 모델링 분석, 2003~2023」. *애니메이션연구*, 20(3), 2024. doi.org/10.51467/ASKO.2024.09.20.3.80
9. (KCI 미등재) **김병준**. 「디지털인문학의 미래, 스마트 빅데이터: 독일 트리어 디지털인문학 센터 CLS 연구 펠로우 후기」. *Digital+Humanities*, 2024. doi.org/10.5281/zenodo.13293696
10. 허예슬*, **김병준***, 최주찬, 최진석. 「푸코의 초상: 한국 현대문학 학술장의 푸코 인용 양상 변화, 2008~2021」. *사회와 이론*, 46, 2023. doi.org/10.17209/st.2023.11.46.305
11. **김병준**, 전봉관. 「민족, 국민, 국가 — 시계열 워드 임베딩을 활용한 조선일보 기사의 민족 담론 의미 변동 추적 (1920~40)」. *현대소설연구*, 90, 2023. doi.org/10.20483/JKFR.2023.06.90.5
12. (우수등재) 전준, **김병준**, 김재홍, 김란우. 「전산사회과학 연구과정의 블랙박스 열기: 아카데믹 데이터베이스를 활용한 비교사회학 연구를 중심으로」. *한국사회학*, 57(2): 131–157, 2023. doi.org/10.21562/kjs.2023.05.57.2.131
13. 최진석*, **김병준***, 허예슬, 최주찬, 황호덕. 「김윤식과 우리 시대, 인용의 인구사회학적 시좌: 현대문학연구자의 성별 및 세대별 김윤식 저술 인용 양상 연구(2004~2019)」. *국제어문*, 96, 2023. doi.org/10.31147/IALL.96.12
14. Yang, J., **Kim, B.**, & Lee, W. (2022). Media coverage of the conflicts over the 4th Industrial Revolution in the Republic of Korea from 2016 to 2020: A text-mining approach. *Asian Journal of Innovation and Policy*, 11(2), 202–226. https://doi.org/10.7545/ajip.2022.11.2.202
15. 서재현, **김병준**, 김민우, 박소정. 「멀리서 읽는 "우리" — Word2Vec, N-gram을 이용한 근대 소설 텍스트 분석」. *대동문화연구*, 115, 2021. doi.org/10.18219/ddmh..115.202109.349
16. **김병준**, 천정환. 「박사학위 논문(2000~2019) 데이터 분석을 통해 본 한국 현대문학 연구의 변화와 전망」. *상허학보*, 60, 2020. doi.org/10.22936/sh.60..202010.012
17. 전성규, **김병준**. 「디지털인문학 방법론을 통한 『서북학회월보』와 『태극학보』의 담론적 상관관계 연구」. *개념과 소통*, 23, 2019. doi.org/10.15797/concom.2019..23.005

18. **김병준**, 전봉관, 이원재. 「비평 언어의 변동: 문예지 비평 텍스트에 나타난 개념단어의 변동 양상, 1995~2015」. *현대문학의 연구*, 61, 2017. doi.org/10.35419/kmlit.2017..61.002
19. 이원재, **김병준**, 전봉관. 「작가-비평가 관계와 비평가의 구조적 위치가 소설 단행본 판매량 증감에 미치는 영향: 2010~2015」. *한국현대문학연구*, 48, 2016. KCI: ART002106457
20. 전봉관, **김병준**, 이원재. 「문예지를 매개로 한 한국 소설가들의 사회적 지형: 1994~2014」. *현대소설연구*, 61, 2016. KCI: ART002092223

## OTHER SCHOLARLY PUBLICATIONS (ROUNDTABLES, REPORTS & ESSAYS)

1. **김병준**. 「AI 시대, 학술 공유지를 지키는 법: 연구자 데이터 주권과 오픈액세스」. *국어국문학*, 215: 327–339, 2026. doi.org/10.31889/kll.2026.6.215.327
2. **김병준**, 유인혁, 김화선, 최성민, 허민석, 배하은, 김완종. 「분업과 연대, 학술 공공성의 새로운 형식들: 한국현대문학 편집인의 열린 대화 (좌담)」. *한국학*, 48(1): 107–131, 2025. doi.org/10.25024/ksq.48.1.202503.107
3. **김병준**. 「학위논문 서지 데이터 분석을 통해 본 한국 현대문학 연구의 변화와 전망」. *상허학보*, 74, 2025. doi.org/10.22936/sh.74..202506.016
4. 최진석, **김병준**, 서희경, 이융희, 이혜령, 홍덕구. 「〈특집〉 제1회 한국현대문학자 대회 — 좌담: 학회란 무엇인가」. *사이間 SAI*, 2024. KCI: ART003087807
5. **김병준**. 「계량서지학 기반 개념사 연구의 가능성」. *개념과 소통*, 28, 2021. KCI: ART002803884

## BOOKS & EDITED VOLUMES

1. 이하람, 김희진, 강수진, 김용수, **김병준**, 박언주, 설연지, 윤미선, 장성현, 전보미, 정서현. 『디지털 인문학의 길잡이: 입문』. 서울대학교출판문화원, 2026. ISBN 9788952173997.
2. 고부열, **김병준**, 남지만, 박철우, 신원철, 안광호, 전현주, 정진한, 최형근, 허인영, 허철. 『동·서 디지털 인문학, 구축·활용·번역』. 동문연, 2026. ISBN 9791124374047. [안양대학교 HK+ 동서교류문헌총서 7]
3. 양소은, 최석영, **김병준**. 『생성형 AI 활용 격차와 AI 리터러시』. 한국언론진흥재단, 2025.
4. **김병준**. 「디지털 인문학: 한국 학술 데이터의 새로운 지평」. 『한국학의 현재와 미래』, 이영준, 김동규, 오영진 엮음. 민음사, 2025.
5. 전봉관, 장우리, 이서준, **김병준**. 『멍텅구리: 한국 최초 신문 연재 네컷만화로 100년 전 날것의 식민지 조선을 보다』. 더숲, 2024. ISBN 9791194273080.
6. 류인태, 곽지은, 권기성, 김바로, **김병준**, 김지선, 박진호, 양승목, 이민철, 이재연, 장문석, 지영원, 한희연. 『디지털로 읽고 데이터로 쓰다: 디지털 한국어문학의 모색』. 박문사, 2023. ISBN 9791192365268.
7. 이재열, 하상응, 임동균, 이원재, **김병준**, 조은아, 강정한, 이호영, 한준. 『플랫폼 사회가 온다: 20대와 50대의 단절과 전승』. 한울아카데미, 2021. ISBN 9788946080744.
8. 김장현, 김윤환, 이겨레, **김병준**. 『데이터기반 디지털 사회과학: 교육과 연구사례』. 성균관대학교출판부, 2019. ISBN 9791155503850.
9. 박건철, 김장현, **김병준**, 이겨레. 『도시데이터표준분석모델: 민원분석편』. 서울디지털재단, 2019. ISBN 9791196601584.

## DATASETS

1. Kim, G., Park, S., Ji, H., Lee, H., Lee, B., Jeong, C., Lee, J.-y., Jo, G., Lim, I., Ismayilov, O., and **Kim, B.** (2026). KNoTE (Korean Novel TEI-Encoded) dataset, v4 [Data set]. Zenodo. doi.org/10.5281/zenodo.19813475
2. Lim, I., Ji, H., Koo, S., Jung, S.-y., Yun, J., and **Kim, B.** (2025). KPoEM (Korean Poetry Emotion Mapping) dataset, v4 [Data set]. Zenodo. doi.org/10.5281/zenodo.15598092

## GRANTS & FUNDED PROJECTS

- **(PI)** "전통문화유산 애니메이션 활용을 위한 한국 전통문화유산 데이터 조사·정리 (Survey and Curation of Korean Traditional Cultural Heritage Data for Use in Animation)", 연구용역 (Commissioned Research), 문화유산기술연구소 (Technology Research Institute for Culture & Heritage, TRIC), 총 연구비 1,200만원 (Total 12 million KRW), 10/2026 – 12/2026.
- **(PI)** "<고백(告白) 벤치마크>: 식민지 남성 지식인 페르소나의 자기합리화 평가 데이터셋 구축 (The Gobaek (Confession) Benchmark: An Evaluation Dataset for Self-Rationalization of Colonial Male Intellectual Personas)", 2026년 2학기 연구교육연계과제, 한국학중앙연구원 (The Academy of Korean Studies), 총 연구비 1,600만원 (Total 16 million KRW), 09/2026 – 02/2027, Link
- **(PI)** "한국 인문학 학술 지형도 구축을 위한 데이터 기반 연구동향 분석 체계 개발 (Developing a Data-Driven Research Trend Analysis Framework for Mapping the Scholarly Landscape of Korean Humanities)", 한국연구원 (Research Institute for Korean Studies), 총 연구비 900만원 (Total 9 million KRW), 04/2026 – 09/2026.
- **(PI)** "문학적 횡단의 지식 그래프: 링크드 오픈 데이터를 활용한 유럽-한국 문학 번역사 모델링 (Knowledge Graphs of Literary Crossings: Modeling European–Korean Literary Translation History Using Linked Open Data)", 한-독 R&D 네트워크 프로그램, 한국연구재단 (National Research Foundation of Korea), 총 연구비 2,000만원 (Total 20 million KRW), 03/2026 – 02/2027, Link
- **(PI)** "인문 데이터 설계 교육을 위한 한국 근현대 문학 TEI/XML 구축 방법론 연구 (TEI/XML Construction Methodology for Korean Modern Literature in Humanities Data Design Education)", 2025년 2학기 연구교육연계과제, 한국학중앙연구원 (The Academy of Korean Studies), 총 연구비 1,600만원 (Total 16 million KRW), 09/2025 – 02/2026, Link
- **(PI)** "고급 자연어 처리 및 대규모 언어 모델 기반 디지털 한국학 연구 및 교육 방법론 개발 (Development of Advanced Natural Language Processing and Large Language Model-Based Digital Korean Studies and Education Methodology)", 2025년 1학기 연구교육연계과제, 한국학중앙연구원 (The Academy of Korean Studies), 총 연구비 1,600만원 (Total 16 million KRW), 03/2025 – 08/2025, Link
- **(Co-PI)** "인공지능과 사회 공진화 동역학에 대한 전산사회과학 연구 (Exploring the Co-evolutionary Dynamics of Artificial Intelligence and Society: A Computational Social Science Approach)", 글로벌인문사회융합연구지원사업(연구그룹형_국외형), 한국연구재단 (National Research Foundation of Korea), 총 연구비 6.6억원 (Total 660 million KRW), 06/2024 – 05/2027.
- **(PI)** "Integrative Approaches in Computational Literary Studies and Global Knowledge Structures", CLS INFRA TNA Fellowships, European Union's Horizon 2020 research and innovation programme, Total 7,240 EUR, 03/2024 – 05/2024, Link · Certificate
- **(PI)** "한국어 인공지능 고도화를 위한 인문사회과학 학술 말뭉치와 언어 모델 구축 (Building an Academic Corpus and Language Model in the Humanities and Social Sciences to Advance Korean AI)", 창의·도전연구

기반지원, 한국연구재단 (National Research Foundation of Korea), 총 연구비 1.4억원 (Total 140 million KRW), 06/2023 – 05/2025, Certificate

## FORTHCOMING & WORKING PAPERS

- "Who Carries the Theory?: A Computational Analysis of Conceptual Travel between Korean and English Literary Studies in South Korea." Invited article for a special issue of *Journal of Cultural Analytics* (Scopus). Forthcoming. Corresponding author.
- "Peripheralizing Social Science Academia: A Relational Landscape of the Global and Local Social Science." Preparing for resubmission. Co-first author.
- "Fine-Tuned Embeddings in Historical RAG: Benchmark Performance and Expert Evaluation on the Veritable Records of the Chosŏn Dynasty." Under review at a Scopus-indexed journal. Co-corresponding author.
- 「문학 데이터로 읽는 서술기계의 변화 — 배명훈의 '서술기계 3부작'(2006~2023)에 나타난 비/인간 주체성」. Under review at a KCI-indexed journal. Corresponding author.
- "Holding the Villain Together: Curated Structure as Authoring Infrastructure for Interactive Characters." Invited to a special issue of an A&HCI-indexed journal (abstract accepted); full manuscript in preparation. Corresponding author.

## SELECTED CONFERENCE PRESENTATIONS

- **Kim, B.**, Ji, H., Kim, G., Jeong, C., Lee, H., & Park, S. (2026, July). Building a Korean literature TEI/XML database: From data acquisition to AI-enhanced reading [Workshop]. ADHO Digital Humanities Conference (DH2026), Daejeon, South Korea. doi.org/10.5281/zenodo.21830474
- 김가연, 박선영, **김병준**. 「2008~2024 KCI 데이터로 본 한국문학/한국영문학 담론의 시계열적 비교 분석」. 제3회 한국현대문학자대회, Feb. 2026. doi.org/10.5281/zenodo.18738333
- Lim, I., & **Kim, B.** (2026, January). Co-Reading: A human–AI co-reader media system for poetry–emotion–color [Paper presentation]. Hong Kong Association for Digital Humanities Conference (HKADH2026), Hong Kong. https://doi.org/10.5281/zenodo.18753718
- **Kim, B.**, & Kim, Y. (2025, July). Digital intellectual history of modern Korean literary studies: Bibliometric analysis of KCI and OpenAlex data sets [Short Paper]. ADHO Digital Humanities Conference (DH2025), Lisbon, Portugal. https://doi.org/10.5281/zenodo.16404466
- Lim, I., Ji, H., & **Kim, B.** (2025, April). Decoding the poetic language of emotion in Korean modern poetry: Insights from a human-labeled dataset and AI modeling [Paper presentation]. Digital Humanities and Social Sciences Korea Conference, Philadelphia, PA, United States.
- **김병준**. 「AI 대전환, 오픈액세스 학술 생태계에 가져올 변화」. 2024 Open Access Conference, Seoul, Oct. 2024 (oral).
- **Kim, B.**, & Jeon, J. (2024, May). Varieties of Korean studies, now and then: A digital intellectual history of Korean studies, 2000–2023 [Paper presentation]. Keimyung International Conference on Korean Studies (KICKS), Daegu, South Korea.
- **Kim, B.** (2024, May). OpenAlex database for computational literary studies and digital intellectual history [Invited talk]. TCDH Research Colloquium, Trier, Germany.

- **Kim, B.**, & Jun, B. G. (2023, June). Quantitative methodology on the success factors of Squid Game based on worldwide audience responses [Paper presentation]. 31st AKSE Conference, Copenhagen, Denmark.

*Additional conference presentations are listed at* byungjunkim.com/talks *and on* Google Scholar.

## TEACHING EXPERIENCE

- Exploring Humanities Data Design, Cultural Informatics, The Academy of Korean Studies (Fall 2025).
- Introduction to Digital Humanities: Foundations of Methodology, Cultural Informatics, The Academy of Korean Studies (Fall 2025).
- Methodology of Digital Korean Language and Literature Research 2, Graduate School, Dept. of Korean Language and Literature, Sungkyunkwan University (Spring 2025).
- Methodology of Digital Content Compilation in Foreign Languages, Cultural Informatics, The Academy of Korean Studies (Spring 2025).
- Research on Humanities Information Compilation 1, Cultural Informatics, The Academy of Korean Studies (Spring 2025).
- Computer Programming for Humanities Knowledge, Cultural Informatics, The Academy of Korean Studies (Fall 2024).
- Social Network Analysis for Digital Humanities, Hallym University (2024).
- Introduction to Digital Humanities II: Computational Literary Studies, Korea University (Fall 2023; selected as a Best Course).
- Literature as Data: The Potential of Quantitative Text Analysis, Winter Institute in Digital Humanities, Korea (Jan. 2023).
- Digital Humanities and Natural Language Processing courses, Sungkyunkwan University (2020–2021).
- Digital Humanities courses, Hankuk University of Foreign Studies (Spring 2021); AI and VR liberal arts courses, Baekseok University (2020–2021).

## TECHNICAL SKILLS & METHODS

- **Programming and data analysis:** Python, R, SQL; reproducible, version-controlled data workflows.
- **NLP and machine learning:** topic modeling, word embeddings, text classification, deep learning, LLM/RAG pipelines.
- **AI agent engineering:** LLM-based agents, tool use, and agentic research workflows.
- **Bibliometrics and network analysis:** KCI, OpenAlex, citation and reference data; Gephi, networkx, igraph, VOSviewer, CiteSpace.
- **Knowledge organization and data infrastructure:** corpus and dataset construction, metadata and TEI/XML, Linked Open Data (RDF/SPARQL), RDBMS, NoSQL.
- **Visualization:** statistical graphics, network visualization, data storytelling for the humanities and social sciences.

## PROFESSIONAL SERVICE

- Local Organizer and Technical Support Committee Chair, ADHO Digital Humanities Conference 2026 (DH2026), Daejeon, 2025 – 2026.
- Co-President, Knowledge Sharing Alliance (지식공유연대; open-access advocacy coalition of Korean scholarly societies), 2019 – Present.
- Board of Directors and Editorial Board, Korean Association for Digital Humanities (KADH), 2022 – Present.
- Operating Committee Member, Korean Modern Literature Scholars' Conference, 2023 – Present.

## AWARDS & HONORS

- 2021 SKKU Graduate Student Paper Award, Sungkyunkwan University (Oct. 2021).
- Best Paper Award, 2021 National Graduate Student Conference, Korean Society for Journalism and Communication Studies (Aug. 2021).
- BK21+ Scholar, National Research Foundation of Korea (2019–2020); Graduate Scholarship, Sungkyunkwan University (2018–2020); National Scholar, KAIST (2015–2017).

## PATENTS

- Kim, G., **Kim, B.**, Woo, D., Park, J., & Lee, S. (2026). System, retrieval method, and index construction method for cross-lingual retrieval of chronicle-style historical records [편년체 사료의 교차언어 검색을 위한 시스템, 검색 방법 및 색인 구축 방법]. Korean Patent Application No. 10-2026-0160921, filed 2026 (pending).
- Kim, G., **Kim, B.**, Woo, D., Park, J., & Lee, S. (2026). Method and apparatus for training cross-lingual encoder based on the correspondence between title and body of chronological historical records [편년체 사료의 제목과 본문의 대응관계에 기초한 교차언어 인코더 학습방법 및 그 장치]. Korean Patent Application No. 10-2026-0160922, filed 2026 (pending).

## REFERENCES

**Dr. Jang Hyun Kim**, Professor of Data Science, Sungkyunkwan University (doctoral advisor). alohakim@skku.edu

**Dr. Wonjae Lee**, Professor of Sociology, KAIST (master's advisor). wnjlee@kaist.ac.kr

**Dr. Bong Gwan Jun**, Professor of Korean Literature, School of Digital Humanities and Computational Social Sciences, KAIST (master's advisor). junbg@kaist.ac.kr